

Measure of Central Tendency & Dispersion

Measure of Central Tendency

A summary measure that attempts to describe a whole set of data with a single value that represents the middle or center of its distribution.

Mean

The mean is the arithmetic average of a data set.

$$\mu = (\sum x_i) / N$$

Where, μ = mean, x_i = datapoint, N = number of datapoints

Mean = sum of observations / number of observations.

Example: 18, 22, 25, 20, 15, 30.

Sum = (18+22+25+20+15+30) = 130; N = count of datapoints = 6;

$\mu = 130 / 6 = 21.67$.

Interpretation: mean value is 21.67.

Grouped Mean

$$\mu = \sum(f_i \times m_i) / \sum f_i$$

Where, μ = mean, f_i = frequency, m_i = midpoint = (Lower Limit + Upper Limit) / 2

Order Fulfillment Duration (Days)	Number of Orders (f_i)	Class Midpoint (m_i)	$f_i \times m_i$	$\mu = \sum(f_i \times m_i) / \sum f_i$
0.0 - 2.0	200	1.0	200.0	= 3300 / 1000
2.0 - 4.0	500	3.0	1500.0	
4.0 - 6.0	250	5.0	1250.0	
6.0 - 8.0	50	7.0	350.0	
	$\sum f_i = 1000$		$\sum f_i \times m_i = 3300$	3.3 days per order

On average, it takes 3.3 days to fulfill an order across the entire dataset of 1,000 orders.

Standard continuous intervals follow a half-open structure ($a \leq x < b$), making lower limits inclusive and upper limits exclusive, so an observation matching an intermediate upper boundary automatically rolls into the next class. To maintain dataset integrity, the upper limit of the final class is made inclusive ($a \leq x \leq b$), ensuring maximum values are fully captured without creating an artificial extra interval.

Median

The Median is the middle value of the sorted data.

Median = datapoint at position $(N + 1) / 2$

where, N = Number of datapoints

Example: 18, 12, 25, 21, 40

Sorted data: 12, 18, 21, 25, 40; N = number of data points = 5

Median Position = $(5+1)/2 = 3$

Median = datapoint at position 3 = 21

Example: 18, 12, 25, 21, 90, 40

Sorted data: 12, 18, 21, 25, 40, 90; N = number of data points = 6

Median Position = $(6+1)/2 = 3.5$

Interpolation: Value_{3.5} = Value₃ + Fraction part of the median position * (Value₄ - Value₃)

Interpolation: $21 + 0.5 * (25 - 21) = 21 + 2 = 23$

Grouped Median

$$\text{Median} = L + [((N/2) - cf_p) / f_m] \times h$$

Where:

L = lower continuous boundary of the median class

N = number of datapoints

cf_p = cumulative frequency of the class preceding the median class

f_m = frequency of the median class

h = class width of the median class = upper boundary - lower boundary

Inclusive Class Interval	Continuous Class Boundaries	Number of Orders (f_i)	Cumulative Frequency (cf)
1-5	0.5 - 5.5	450	450
6-10	5.5 - 10.5	300	750
11-15	10.5 - 15.5	150	900
16-20	15.5 - 20.5	100	1,000 (N = 1,000)

The Median Class is the class interval where the cumulative frequency first meets or exceeds half the total observations: $cf \geq N/2$

Median class is 5.5 - 10.5:

Median = $5.5 + [(1000/2 - 450) / 300] * 5 = 6.33 \text{ items}$

50% of all orders contain 6.33 items or fewer, and 50% contain 6.33 items or more.

Mode

The mode is the most frequent value or category in a dataset.

Cases of the Mode:

- **Unimodal (One Mode):** A dataset where exactly one value appears with the highest frequency.
Example: Card, Card, Wallet, Card, Cash, Wallet
Frequencies: Card -> 3, Wallet -> 2, Cash -> 1
Mode: Card
- **Bimodal (Two Modes):** A dataset where two distinct values share the exact same highest frequency.
Example: Card, Cash, Wallet, Card, Cash
Frequencies: Card -> 2, Cash -> 2, Wallet -> 1
Mode: Card and Cash

- **Multimodal (More Than Two Modes):** A dataset where three or more values tie for the highest frequency.
Example: Card, Card, Cash, Cash, Wallet, Wallet, UPI
Frequencies: Card -> 2, Cash -> 2, Wallet -> 2, UPI -> 1
Mode: Card, Cash, and Wallet
- **No Mode:** A dataset where every value appears with the exact same frequency (no single value or subset of values is more frequent than the others).
Example: Card, Cash, Wallet, Card, Cash, Wallet
Frequencies: Card -> 2, Cash -> 2, Wallet -> 2
Mode: No mode

Example:

Is_Returned: 1, 1, 1, 0, 0, 1, 0, 0, 0

mode: 0

Grouped Mode

$$\text{Mode} = L + \left[\frac{(f_m - f_1)}{(2f_m - f_1 - f_2)} \right] \times h$$

Mode Class: The Modal Class is the class interval containing the highest individual frequency (f_m).

where,

L = lower continuous boundary of the modal class

f_m = frequency of the modal class

f_1 = frequency of the class preceding the modal class

f_2 = frequency of the class succeeding the modal class

h = class width of the modal class = upper boundary - lower boundary

Inclusive Class Interval	Continuous Class Boundaries	Number of Orders (f_i)	Mode
1-3	0.5 - 3.5	120	$f_m = 450$ $f_1 = 120$ $f_2 = 280$ $L = 3.5; h = 3$ 5.48 Hours
4-6	3.5 - 6.5	450	
7-9	6.5 - 9.5	280	
10-12	9.5 - 12.5	150	

The order processing duration that occurs most frequently (the single most common processing time) is 5.48 hours.

Measure of Dispersion

It is a summary value that describes how spread out or scattered the data points are around the center.

- **Range:** The difference between the maximum (highest) and minimum (lowest) values in a dataset. It measures the total spread across the entire dataset.

$$\text{Range} = X_{\max} - X_{\min}$$

- **Variance:** The average of the squared deviations from the mean. It quantifies how far individual data points fall from the center on average, measured in squared units.

$$\sigma^2 = \sum (x_i - \mu)^2 / N$$

- **Grouped Variance:**

$$\sigma^2 = \sum [f_i \times (x_i - \mu)^2] / \sum f_i$$

- **Standard Deviation:** It measures the average distance of data points from the mean in the original units of the data.

$$\sigma = \sqrt{\sigma^2}$$

- **Coefficient of Variance:** It represents the relative percentage of variation present in a dataset compared to its average value.

$$CV = (\sigma / \mu) \times 100$$

Ungrouped Dispersion Example

Duration (Days)	μ	Range	$x_i - \mu$	$(x_i - \mu)^2$	σ^2	σ	CV
2.0			-1.1	1.21			
3.5			0.4	0.16			
1.5			-1.6	2.56			
4.0			0.9	0.81			
2.5			-0.6	0.36			
5.0			1.9	3.61			
3.0			-0.1	0.01			
2.0			-1.1	1.21			
4.5			1.4	1.96			
3.0			-0.1	0.01			
N = 10	3.1	3	$\sum (x_i - \mu)^2 : 11.9$	11.9	1.19	1.09	35.16%

A Standard Deviation of 1.09 days shows that fulfillment times typically fluctuate by ± 1.09 days around the average (3.1 days), while a CV of 35.16% indicates moderate relative volatility.

Grouped Dispersion Example

Class (Days)	Orders (f_i)	Midpoint (x_i)	$f_i \times x_i$	μ	Dev ($x_i - \mu$)	Sq Dev ($(x_i - \mu)^2$)	Wtd Sq Dev ($f_i \times (x_i - \mu)^2$)	σ^2	σ	CV
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0.0-2.0	200	1.0	200.0		-2.3	5.29	1058.0			
2.0-4.0	500	3.0	1500.0		-0.3	0.09	45.0			
4.0-6.0	250	5.0	1250.0		1.7	2.89	722.5			
6.0-8.0	50	7.0	350.0		3.7	13.69	684.5			
Total	1000		3300.0	3.3			2510.0	2.51	1.58	47.88%

A Standard Deviation of 1.58 days shows that typical fulfillment times vary by 1.58 days around the average, while a CV of 47.88% indicates moderate process volatility.

Histograms and Their Usage

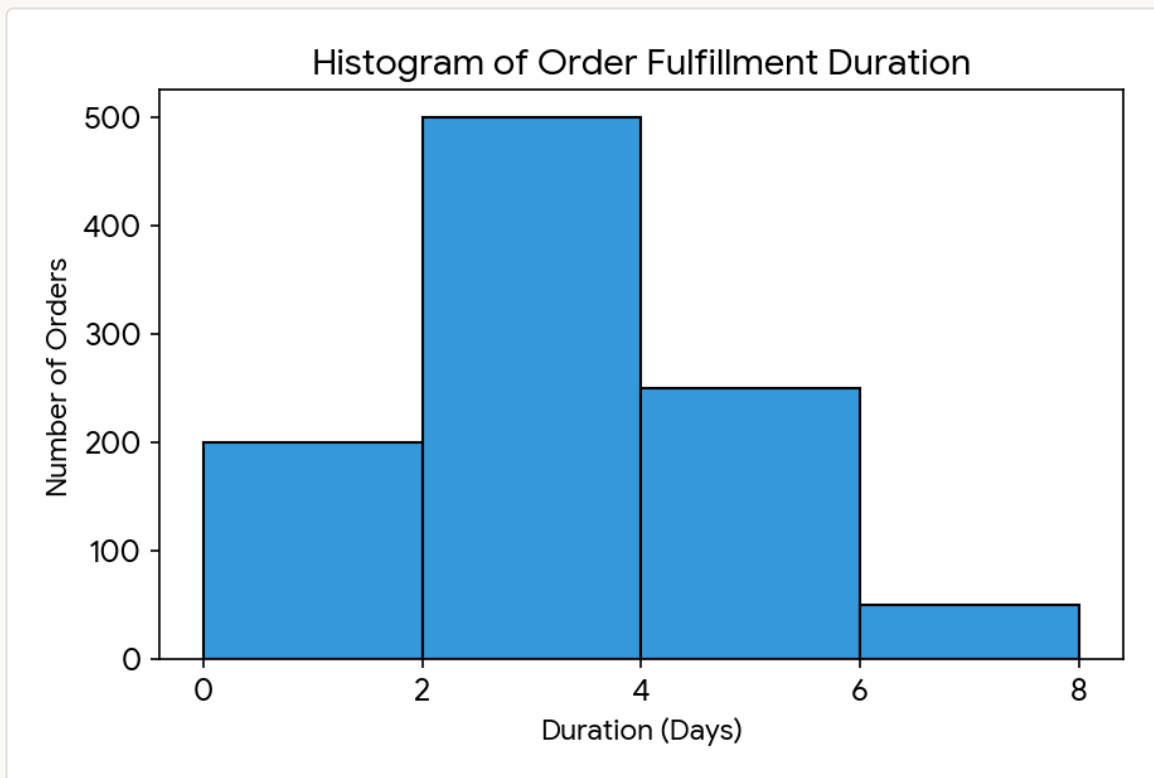
What is a Histogram?

A **histogram** is a visual representation of continuous, grouped data (like the "Continuous Class Boundaries" tables above). It looks similar to a bar chart, but with key differences:

- The x-axis represents continuous intervals or class boundaries (e.g., 0.0 - 2.0, 2.0 - 4.0).
- The y-axis represents the frequency (the number of data points, like the "Number of Orders").
- Because the data represents continuous ranges, the bars in a histogram touch each other with no gaps in between, directly illustrating the flow from one continuous class to the next.

Visual Example

Below is a proper histogram plotted using the "Order Fulfillment Duration" Grouped Dispersion Example from above. Notice how the bars touch precisely at the continuous class boundaries (0, 2, 4, 6, 8) without any gaps, differentiating it from a standard column chart.



Uses of a Histogram

Visualizing data with a histogram serves several practical purposes that raw numbers and tables cannot easily convey:

- **Identifying the Mode Visually:** In a histogram, the tallest bar instantly reveals the Modal Class (the most frequent range of data) without needing to look at a frequency table.
- **Understanding Data Shape:** It immediately shows if the data is symmetrical (bell-shaped) or asymmetrical, revealing the underlying distribution pattern.
- **Spotting Outliers:** If there are isolated bars sitting far away from the main cluster of bars, those represent anomalies or extreme outliers in the dataset.

- **Grasping Dispersion:** A wide, flat histogram visualizes a high variance and Standard Deviation, while a narrow, tall histogram shows low variance (meaning the data is clustered tightly around the mean).